

Larynx and Voice

Laryngeal and head and neck cancer (alcohol, tobacco and HLA), laryngopharyngeal reflux, pharyngeal diseases and the genetics of voice pitch

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F's genes (genotyping chip + local imputation). It is not a diagnosis or a prescription. All actions are marked as suggestion (not applied).

Contents

Section	Page
1. Summary	3
2. How to read this report	4
3. Polygenic scores	5
4. Key variants by topic	6
5. GWAS loci burden	8
6. Favourable	9
7. Unfavourable	10
8. Mechanisms of ageing	11
9. Practical suggestions	12
10. Limitations	13
11. Methods and sources	14
12. Annex: all variants analysed	15

1. Summary

Favourable. **ALDH2*2** is absent (G/G), so acetaldehyde from alcohol does not accumulate (and alcohol is at zero). The genomic scores for head and neck squamous cell carcinoma are at percentiles 70.0 and 46.1, and those for the mouth and oropharynx at percentiles 39.3 and 50.9; the consensus is **48.5** (typical) and the loci burden for these cancers is at percentile 31.0. Pharyngeal and tonsil diseases have a loci burden at **2.8**, with LTBR lacking the risk allele.

Unfavourable. The protective **ADH1B*2** is absent (C/C) and **ADH7** has 2 risk alleles (C/C): these variants only matter with alcohol. **CHRNA5 D398N** has 1 allele (G/A), which only matters with tobacco. There is 1 allele in the HLA loci for head and neck and oropharyngeal cancer (linked to HPV) and in TERT/CLPTM1L. For reflux, the main cause of chronic laryngitis, the scores disagree: the 2024 genomic score is at percentile **83.1**, the 2022 score at percentile 6.8, and the loci burden at percentile 75.0.

What it means. The genetic risk of laryngeal and head and neck cancer is typical and almost entirely conditional on alcohol and tobacco, both at zero. The larynx-specific scores have very low coverage and cannot be interpreted. The voice and the larynx depend mainly on voice use, hydration and reflux. The genetics of voice pitch is descriptive only.

2. How to read this report

- **Data:** genotypes from Sérgio F's chip (GRCh38, + strand) and, when the SNP is not on the chip, locally imputed genotypes (Beagle 5.5, public panel of 30x whole genomes) with quality $DR2 \geq 0.8$. Below this threshold the SNP is treated as not found in the data.
- **Key variants:** 12 SNPs selected from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 9 from the chip, 3 imputed and 0 not found.
- **Strand:** all genotypes are on the + strand of GRCh38. The published effect allele is compared with the observed alleles; 1 SNP needed conversion to the complementary strand and 2 are palindromic (A/T or C/G), resolved by the reference allele and the position. Where the literature uses the gene strand, the letters may differ from those in this report.
- **Polygenic scores (PGS):** 10 PGS Catalog scores calculated locally with 5,636,609 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$), calculated with exactly the same variants. Percentile ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **GWAS loci burden:** for each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in the GWAS Catalog (1 row per SNP, the most significant) are added up as a simple count of effect alleles (each allele counts 1). Sérgio F's sum is compared with that of the reference population ($n = 633$) using exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium with each other, so the burden is a complement to, not a substitute for, the polygenic scores. 195 loci, 142 with genotype.
- **Judgement:** each result is classed as Favourable or Unfavourable (or typical / no judgement). A genotype is a probability, not a destiny; almost all effects are small and add to lifestyle.
- **Evidence:** Strong = replicated and with a clear clinical effect; Moderate = consistent GWAS/meta-analyses, modest effect; Weak = small or purely mechanistic studies.

3. Polygenic scores

Consensus by trait. For each group, the consensus is the **median of the percentiles of the scores with effect-size coverage $\geq 75\%$** , all calibrated in the same reference population ($n = 633$). The genomic column is the median of the scores with $\geq 100,000$ variants. Scores with coverage $< 75\%$ are shown in the tables but do not enter the consensus.

Group	Consensus (percentile)	Scores with coverage $\geq 75\%$	Genomic scores (percentile)	Genomic scores
Laryngeal cancer	—	0 of 3	—	0
Head and neck	48.5	4 of 5	48.5	4
Reflux (link to the Oesophagus report)	44.9	2 of 2	44.9	2

In all the cancer and reflux scores, a higher percentile means more risk. The 3 laryngeal cancer scores (2020) have 11–22% coverage and receive no judgement; the genomic head and neck scores (2023 and 2026), with 97% coverage, replace them.

Laryngeal cancer

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Laryngeal cancer (genomic)	PGS000360 (Fritsche 2020, 26,000 variants)	5,235 of 25,920 (coverage 22%)	98.4	MORE LIKELY	no judgement (low reliability)
Laryngeal cancer (genomic)	PGS000361 (Fritsche 2020, 88,000 variants)	16,998 of 87,893 (coverage 20%)	82.3	MORE LIKELY	no judgement (low reliability)
Laryngeal cancer	PGS000362 (Fritsche 2020, 53 variants)	8 of 53 (coverage 11%)	3.8	LESS LIKELY	no judgement (low reliability)

- **Laryngeal cancer (genomic):** Effect-size coverage of 11–22%: the variants of this score (2020) are mostly outside the chip and the good-quality imputation. It does not enter the consensus or the judgement.
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Head and neck

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Head and neck squamous cell carcinoma (genomic, 2026)	PGS012586 (Lee 2024, 972,000 variants)	952,505 of 972,182 (coverage 97%)	70.0	TYPICAL	typical
Head and neck squamous cell carcinoma (genomic)	PGS003390 (Namba 2022, 931,000 variants)	911,178 of 931,477 (coverage 97%)	46.1	TYPICAL	typical
Oral cavity cancer (genomic)	PGS012587 (Lee 2024, 972,000 variants)	952,505 of 972,182 (coverage 97%)	39.3	TYPICAL	typical
Oropharyngeal cancer (genomic)	PGS012588 (Lee 2024, 972,000 variants)	952,505 of 972,182 (coverage 97%)	50.9	TYPICAL	typical
Mouth and pharynx cancer	PGS000081 (Graff 2020, 14 variants)	9 of 14 (coverage 55%)	42.3	TYPICAL	typical

Reflux (link to the Oesophagus report)

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Gastro-oesophageal reflux (genomic)	PGS004468 (Jung 2024, 1.06 million variants)	1,025,639 of 1,059,939 (coverage 96%)	83.1	MORE LIKELY	unfavourable
Oesophagitis, reflux and related diseases (genomic)	PGS002063 (Privé 2022, 836,000 variants)	820,027 of 836,413 (coverage 97%)	6.8	LESS LIKELY	favourable

4. Key variants by topic

Variants in alcohol metabolism (ADH1B, ADH7, ALDH2), nicotine dependence (CHRNA5), the HLA loci of head and neck cancers, reflux, pharyngeal diseases and voice pitch. Published alleles with a protective effect were inverted to the risk direction.

Laryngeal and head and neck cancer

Almost all of the common inherited risk acts through alcohol, tobacco and HPV.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ADH1B	rs1229984	Arg48His	C/C	imputed DR2=0.88	T (0 copies)	neutral	ADH1B*2 (Arg48His): the T allele accelerates ethanol conversion and protects from heavy consumption; without it, with alcohol, there is more cancer of the mouth, pharynx and larynx	Strong
ADH7	rs971074		C/C	chip	G (2 copies)	unfavourable	ADH7 (mucosal alcohol dehydrogenase): mouth and pharynx cancer [GWAS Catalog: Oral cavity and pharyngeal cancer, allele G, OR 1.33, p=9E-17, PMID 21437268]	Moderate
ALDH2	rs671	Glu504Lys	G/G	chip	A (0 copies)	favourable	ALDH2*2 (Glu504Lys): the A allele accumulates acetaldehyde with alcohol (more head and neck cancer)	Strong
CHRNA5	rs16969968	D398N	G/A	chip	A (1 copy)	unfavourable (mild)	D398N of the nicotinic receptor α5: the A allele increases nicotine dependence and cancers linked to tobacco (lung, larynx)	Strong
HLA-DQB1	rs3135001		T/C	imputed DR2=1.00	C (1 copy)	unfavourable (mild)	HLA-DQB1: head and neck squamous cell carcinoma [GWAS Catalog: Head and neck squamous cell carcinoma, allele C, OR 1.30, p=1E-16, PMID 32276964]	Moderate
HLA-DRB1/DQA1	rs34518860		G/A	chip	G (1 copy)	unfavourable (mild)	HLA class II: oropharyngeal cancer (linked to HPV) [GWAS Catalog: Oropharynx cancer, allele G, OR 1.64, p=3E-17, PMID 32276964]	Moderate
TERT/CLPTM1L	rs401681		C/T	chip	C (1 copy)	unfavourable (mild)	5p15.33 (TERT/CLPTM1L): several epithelial cancers, including the larynx [GWAS Catalog: Non-small cell lung cancer, allele C, OR 1.16, p=8E-12, PMID 31326317]	Moderate

Laryngopharyngeal reflux and pharynx

Acid reflux reaching the larynx is the most common cause of chronic hoarseness.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CRTC1	rs10404696	T/T	chip		G (0 copies)	favourable	CRTC1: gastro-oesophageal reflux (reflux is the main cause of chronic laryngitis) [GWAS Catalog: Acid reflux/GERD, allele G, beta 0.041 [0.03-0.052] unit increase, p=7E-15, PMID 39024449]	Moderate
ATXN2/SH2B3	rs4766578	A/A	chip		T (0 copies)	favourable	ATXN2/SH2B3 region: gastro-oesophageal reflux [GWAS Catalog: Esophagitis, GERD and related diseases (PheCode 530.1), allele T, beta 0.032 [0.023-0.04] unit increase, p=1E-15, PMID 39024449]	Moderate
LTBR	rs10849448	G/G	chip		A (0 copies)	favourable	LTBR (lymphotoxin β receptor, pharynx lymphoid tissue): pharyngeal disease [GWAS Catalog: Pharyngeal disease, allele G, OR 0.90; risk/increase allele = A, the opposite, p=2E-22, PMID 36653354]	Moderate
HORMAD2/LIF	rs713875	C/G	chip		G (1 copy)	unfavourable (mild)	HORMAD2/LIF (22q12): pharyngeal and tonsil disease [GWAS Catalog: Pharyngeal disease, allele G, OR 1.08, p=7E-18, PMID 36653354]	Moderate

Voice and vocal folds

Descriptive genetics of voice pitch (no judgement).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ABCC9	rs11046212	C/C		imputed DR2=1.00	T (0 copies)	no judgement	ABCC9 (smooth muscle potassium channel): associated with median voice pitch [GWAS Catalog: Voice pitch (median), allele T, beta 0.110 [0.09-0.14] SD unit increase, p=8E-24, PMID 38714798]	Weak

5. GWAS loci burden

Sum of GWAS Catalog loci ($p \leq 5 \times 10^{-8}$) for head and neck cancers, pharyngeal diseases, reflux and voice pitch. Alleles are oriented towards more risk (or towards a higher pitch, for voice) and the sum is compared with the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F's effect alleles	Calibrated variants	Percentile	Judgement
Head and neck cancer (larynx, pharynx, mouth)	70 (29)	25 of 58	29	31.0	typical
Pharyngeal and tonsil diseases	23 (17)	11 of 34	17	2.8	favourable
Gastro-oesophageal reflux	98 (92)	88 of 184	91	75.0	typical
Voice pitch	4 (4)	4 of 8	4	57.5	no judgement

6. Favourable

- **ALDH2*2 absent** (G/G): no acetaldehyde build-up. **Strong.**
- **Typical head and neck carcinoma:** consensus at percentile **48.5** (4 genomic scores); loci burden at percentile 31.0. **Moderate.**
- **Pharyngeal and tonsil diseases:** loci burden at **2.8**; LTBR without the risk allele (G/G). **Moderate.**
- **CRTC1** (T/T) and **ATXN2/SH2B3** (A/A) without the reflux alleles. **Weak.**

7. Unfavourable

- **ADH1B*2 absent** (C/C) and **ADH7** C/C (2 alleles): more risk only with alcohol, which is at zero. Moderate (conditional).
- **CHRNA5 D398N** G/A: 1 allele; more risk only with tobacco. Strong (conditional).
- **Reflux**: 2024 genomic score at percentile **83.1** versus 6.8 for the 2022 score; loci burden at percentile 75.0. Mixed signal. **Weak.**
- **HLA-DQB1** T/C, **HLA-DRB1/DQA1** G/A (oropharynx, linked to HPV) and **TERT/CLPTM1L** C/T: 1 allele each. **Weak.**

8. Mechanisms of ageing

Mapping onto the 84 ageing mechanisms of Sérgio F's immortality goals (only the mechanisms touched by this report).

Mechanism	Genetic finding	Meaning
DNA Damage and DNA Repair Deficit	Acetaldehyde from alcohol (normal ALDH2; slow ADH1B and ADH7 with 2 alleles): damage to mucosal DNA only with alcohol	Conditional
Somatic Mutation Accumulation	Head and neck cancers with consensus at percentile 48.5; risk dependent on tobacco, alcohol and HPV	Typical
Telomere Attrition and Hayflick Limit	TERT/CLPTM1L (5p15) with 1 allele, shared with several epithelial cancers	Typical
Chronic Inflammation	Laryngopharyngeal reflux with a mixed genetic signal; pharyngeal diseases unlikely	Typical
Immunosenescence	Oropharyngeal HLA loci (response to HPV) with 1 allele; the HPV vaccine protects	Conditional

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Zero alcohol and zero tobacco** (already in the plan): they cancel almost all of the effect of ADH1B, ADH7 and CHRNA5. *Suggestion (not applied).*
2. **Hoarseness lasting more than 3 weeks** (especially without a cold) justifies a laryngoscopy; this is the clinical rule and genetics does not change it. *Suggestion (not applied).*
3. **Laryngopharyngeal reflux**: if there is throat clearing, a feeling of something stuck in the throat or morning hoarseness, the simple measures are dinner 3 hours before bed and raising the head of the bed. CYP2C19 *1/
*1 gives normal efficacy to proton pump inhibitors, if they are needed. *Suggestion (not applied).*
4. **Voice care**: hydration, breaks during prolonged voice use and not shouting over noise. *Suggestion (not applied).*
5. **HPV vaccine**: discuss with the doctor; it protects against HPV-related oropharyngeal cancer. *Suggestion (not applied).*

10. Limitations

- The larynx-specific cancer scores have very low coverage and cannot be interpreted; the reading uses the genomic head and neck scores.
- The reflux scores disagree with each other; see the Oesophagus report.
- The voice depends mainly on use, hydration and reflux; few loci are known.
- Head and neck cancer depends mainly on tobacco, alcohol and HPV; genetics explains a small part.
- The loci burden adds up SNPs in linkage disequilibrium; it is descriptive.
- Conclusions from genes only; this is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F's chip (GRCh38, + strand). Missing SNPs come from local imputation (Beagle 5.5, public panel of 30x whole genomes), used only with $DR2 \geq 0.8$. No genetic data left the analysis computer.
- **Alleles and strand:** the published effect allele is compared with the alleles of the imputed VCF; for palindromic SNPs the + strand is assumed (the GWAS Catalog and PGS Catalog convention). Published alleles with a protective effect were inverted to the risk direction.
- **Polygenic scores:** GRCh38 harmonised files from the PGS Catalog (10 scores: laryngeal, head and neck, oral and oropharyngeal cancer, and reflux). Score = $\sum \text{coefficient} \times \text{dosage}$. The percentile is empirical in the reference population ($n = 633$), using the same variants. Consensus = median of the scores with effect-size coverage $\geq 75\%$.
- **Loci burden:** GWAS Catalog ($p \leq 5 \times 10^{-8}$), 4 sets, alleles oriented by the sign of the published effect, with empirical calibration in the same reference population.
- **Interpretation:** McKay 2011 and Lesseur 2016 (head and neck cancer GWAS), Hashibe 2008 (ADH1B, ADH7 and ALDH2), Thorgeirsson 2008 (CHRNA5), Ong 2022 (reflux), PGS Catalog, GWAS Catalog and ClinVar.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Strand	Interpretation
Laryngeal and head and neck cancer	ADH1B	rs1229984	C/C	imputed DR2=0.88	T	0	+ strand	without the protective allele (common genotype)
Laryngeal and head and neck cancer	ADH7	rs971074	C/C	chip	G	2	complement (– strand)	2 copies of the risk allele
Laryngeal and head and neck cancer	ALDH2	rs671	G/G	chip	A	0	+ strand	without the risk allele
Laryngeal and head and neck cancer	CHRNA5	rs16969968	G/A	chip	A	1	+ strand	1 copy of the risk allele
Laryngeal and head and neck cancer	HLA-DQB1	rs3135001	T/C	imputed DR2=1.00	C	1	+ strand	1 copy of the risk allele
Laryngeal and head and neck cancer	HLA-DRB1/ DQA1	rs34518860	G/A	chip	G	1	+ strand	1 copy of the risk allele
Laryngeal and head and neck cancer	TERT/ CLPTM1L	rs401681	C/T	chip	C	1	+ strand	1 copy of the risk allele
Laryngopharyngeal reflux and pharynx	CRTC1	rs10404696	T/T	chip	G	0	+ strand	without the risk allele
Laryngopharyngeal reflux and pharynx	ATXN2/ SH2B3	rs4766578	A/A	chip	T	0	+ strand (palindromic; + assumed)	without the risk allele
Laryngopharyngeal reflux and pharynx	LTBR	rs10849448	G/G	chip	A	0	+ strand	without the risk allele
Laryngopharyngeal reflux and pharynx	HORMAD2/ LIF	rs713875	C/G	chip	G	1	+ strand (palindromic; + assumed)	1 copy of the risk allele
Voice and vocal folds	ABCC9	rs11046212	C/C	imputed DR2=1.00	T	0	+ strand	0 copies of the effect allele